

Where is the Fire?

The Internet is one of the best things that ever happened to mankind. But like all good things in life, it also comes with its share of woes. What was once meant to be a means for sharing important information across geographical boundaries, has now turned into a vehicle that lets hackers sit in the comfort of their dens and wreak havoc on unsuspecting users. Securing systems and data from such malicious activity has been the top concern of leading corporate houses across the globe, and has led to a billion dollar Internet security industry. Almost all organisations employ the services of professional security companies to secure their networks from hackers. Despite this, there has been an increase in hacking activity and security exploits worldwide.

If this is the case with corporates, then what about home users? Aren't they prone to the same level of vulnerabilities? Well, not really; it's actually much worse. A majority of home users access the Internet without installing any security software in the belief that their computers are too unimportant to be hacked. This is no longer the case (see box, 'Why would Anybody Hack my PC?'), and even if it were, there are enough mischief mongers out there who wouldn't mind spending hours hacking into a PC just for fun.

So what do you do about this threat? Simple, just install one of the personal firewalls that are specifically tailored to protect the home user against crackers. These applications quietly sit on your PC and look through all the tons of traffic that goes through your network connection everyday. They protect you against all the bad guys out there trying to get a hold of your PC.

The internals

A firewall's basic function is to protect your computer against any form of Internet abuse. This is easier said than done for the number of new methods and tools available for hacking is simply mind-boggling. Fortunately (or unfortunately), all these tools are based upon certain age-old exploits that were found in the protocols running the Internet. These include various forms of Port Scanning, Packet filtering and Denial Of Service (DOS) attacks. Firewalls are always standing guard at your PC's doorstep trying to figure out genuine pings (see box on 'Useful Commands') and denying

requests from the dangerous ones. In short, they should have enough intelligence to catch newer tools that use some variations of the older techniques in the same way as anti-virus software have to use heuristics to catch newer strains of older viruses.

Firewalls use three primary techniques to protect you from malicious code and hackers. These are Packet Filtering, State Inspection and Application Level Inspection

In Packet Filtering, the firewall simply inspects the IP addresses of the packets, and compares it with an access control list to decide upon the fate of the concerned packet. Since the firewall does not have any in-built intelligence and it does not remember any previous information about the packets transmitted—the overhead place by this technique is minimum.

The next level of security is provided by State Inspection firewalls. They not only inspect the content of the packet, but also look through their behavioural history to decide whether to grant access or not.

Application Level Inspection is the most advanced technique, whereby every layer of the TCP/IP protocol stack is inspected, and a decision is taken accordingly. Obviously, this level of inspection also places maximum overhead on the system.

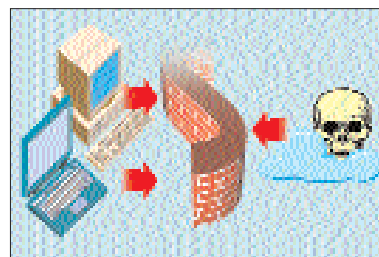
How we tested

The tests were designed to test the firewall's capability for providing basic security, as well as all the enhanced features.

Information Leakage Tests: The firewall's ability to resist external users from gaining information—such the operating system used, number of open ports—about the target machine is tested here.

Helpful commands

Apart from using a firewall, there are some very useful commands that can be used to monitor your Internet connection. One of the most commonly used commands is 'netstat'. It gives a list of the all the open ports on your machine along with the protocols being used, the foreign addresses that are accessing the ports, and the state of each connection. This command can really come in handy at times when you are suspicious of the activities going on your PC. Other commands that can be used to troubleshoot your Internet connection are 'ping' and 'tracert'.



A software firewall sits on your laptop or desktop and scrutinises all Internet traffic



Ports are like doors to your system, and if left ajar, any smart hacker could break into your PC



Packet filtering firewalls only look at the source and destination IP addresses of a packet, and then decide its fate

